

JASH THAKKAR

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EDUCATION

Georgia State University – Atlanta, GA

Bachelor of Science in Computer Science | May 2026 | GPA: 3.5 / 4.0

- Relevant Coursework: Data Structures, Data Science, Algorithms, Systems Programming, Operating Systems, Software Engineering, Software Development, Fundamentals, Calculus I and II, Linear Algebra, Statistics, Mobile Development, Web Programming, Cybersecurity Fundamentals

PL-300 Microsoft Certificate: Power BI Data Analyst Associate | In progress

WORK EXPERIENCE

Freelance

Software Engineer | Georgia | June 2024 – Present

- Developed 6+ full stack web applications for local businesses using **HTML, CSS, JavaScript, PHP, and SQL** delivering scalable, user-friendly, and production ready software.
- Deployed websites to production, using **Vercel** hosting, **MySQL** for database configuration, **Git** based version control, and domain setup.
- Improved product usability, accessibility, and performance, directly contributing to an average of **20%** increase in user engagement through client projects.
- Gathered requirements through conversations with clients and translated business needs to technical solutions demonstrating strong communication and product thinking skills.
- Implemented authentication systems, including flows for logging in and creating accounts, password hashing, session management, and input validation to prevent security risks.

Select Management Resources

Software Engineer Intern | Alpharetta, Georgia | January 2023 – July 2023

- Developed **Python** and **JavaScript** scripts to streamline troubleshooting workflows, reduce repetitive operational tasks, and improve incident response efficiency.
- Designed and maintained structured technical documentation, system guides, and process workflows, improving onboarding efficiency by **20%**.
- Investigated application, system, and network issues through log analysis, ticket data review, and root cause analysis, contributing to a **25%** reduction in recurring incidents.
- Collaborated with infrastructure and operations teams to identify system reliability gaps and implement tooling and process improvements.
- Debugged and resolved complex software and system issues using structured problem solving, scripting-based validation, and logical decomposition.
- Produced internal data driven reports and dashboards to track issue trends and performance metrics, enabling data driven operational decisions.

PROJECTS

IT Ticket Analytics (Python, SQL, PostgreSQL, Power BI, Excel, Pandas, Faker, SQLAlchemy, DAX, pgAdmin)

- Built an end-to-end IT ticket analytics platform using **Python, PostgreSQL, SQL, Excel, and Power BI** to simulate enterprise operational ticket reporting workflows, through synthetic data generation, relational database modeling, analytical querying, automated reporting, and dashboard visualizations.
- Developed and queried a PostgreSQL relational database containing **10,000+** synthetic tickets using advanced SQL techniques including joins, aggregations, filtering, KPI calculations, indexing, constraints, and reusable reporting views for SLA compliance, backlog aging, team performance, and impact analysis.
- Built automation pipelines using python including libraries like **pandas, Faker, and SQLAlchemy** to generate realistic ticket datasets, load CSV data into PostgreSQL, validate database connections, and automate multi sheet Excel reporting exports.
- Created a multi-page Power BI dashboard connected directly to PostgreSQL, using DAX measures, KPI cards, slicers, relational data modeling, backlog aging analysis, SLA monitoring, team analytics, and executive data visualizations.

A/B Testing Engine (Python, Streamlit, scikit-learn, Pandas, NumPy, SciPy, Plotly)

- Designed an end-to-end experimentation analytics platform that uses real, and simulated A/B testing data to perform statistical significance testing using two proportion z-tests and t-tests to evaluate features.
- Implemented a hybrid modeling system that combines **Logistic Regression and Random Forest** models to predict user conversion probability, compare model performance and generate a prediction using confidence through **ROC-AUC**.
- Engineered ELT style feature data processing pipelines for user behavior data, that included data validation, data cleaning, transformation, aggregation, and feature engineering for statistical analysis and dashboard developments.
- Built a decision engine that uses statistical results and ML predictions to produce a rollout recommendation based on significance, lift, and confidence.
- Developed a Streamlit dashboard and CLI interface to visualize experiment results including conversion rates, confidence intervals, and comparisons between models, giving a user-friendly system for data driven decisions and giving visuals for data modeling through the interactive dashboards.

Traffic Sign Image Recognition System (Python, PyTorch, scikit-learn, OpenCV, Pandas, NumPy, Torchvision, Matplotlib)

- Built an image recognition and classification models for **43** traffic sign classes by training, evaluating, and optimizing multiple machine learning models.
- Developed a **Neural Network** using **ResNet18** and **Transfer Learning** achieving up to a **94%** test accuracy through stage-based training, and fine tuning.
- Implemented multiple baseline models including **HOG + SVC, k-NN, and Random Forest** models, and through hyperparameter tuning was able to improve model test accuracies from as low as **77%** to as high as **97%** improving overall model accuracies by **10%-20%**.
- Evaluated models based statistical analysis using precision, recall, F1-score, and matplotlib for confusion matrices visualization to analyze class level errors to better tailor optimizations and improve accuracies
- Engineered data preprocessing, data cleaning, and augmentation pipelines, including normalization and standardization of the image features.

TECHNICAL SKILLS

- Programming Languages: Python, Java, R, JavaScript, C++, C, C#, Dart, HTML, CSS, SQL, PHP, YAML
- Technologies: Power BI, Tableau, Excel, GitHub, Git, Microsoft Office, Google Suite, MySQL, PostgreSQL, AWS, DBeaver, Vercel, Linux, Unix, Command Line, Data Structures, Algorithms, PyTorch, scikit-learn, Matplotlib, Pandas, NumPy, Faker, SQLite, SQLAlchemy, DAX, pgAdmin
- Core Skills: Object Oriented Programming, Database Management, Debugging, Troubleshooting, Automation Scripting, Server-Side Programming, Version Control, Prompt Engineering, Machine Learning, Data Processing, Data visualization, Data Analysis, Business Intelligence, Dashboard Development